

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_MonteCarlo")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
#seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

# text_priors <- '
# Distrib (pAm,          TruncNormal,          1172,          117.2, 800, 8000);
# Distrib (r_pAm_pM,    TruncNormal,          1.5,          0.2, 0.01, 100);
# Distrib (Fm,          TruncNormal,          0.3,          0.1, 0.1, 0.6);
# Distrib (kap,         Uniform,              0.05,          0.25);
# Distrib (Ehb, TruncNormal, 0.5, 0.5, 0.1, 10);
# Distrib (Ehp, TruncNormal, 100, 50, 1, 1000);
# # Distrib (alpha_Ehb, TruncNormal,          0.8,          0.1, 0, 1);
# # Distrib (beta_Ehp,  TruncNormal,          10,          1, 1, 1000);
# Distrib (kJ,          LogUniform,          0.00001,          0.1);
# Distrib (Eg,          TruncNormal,          6880.33,          1000, 0, 30000);
# Distrib (E_coc,       TruncNormal, 200, 100, 50, 5000);
#
# Distrib (muOM,        Normal,              11700,          500);
# Distrib (rOM_ClxHorse, TruncNormal,          0.2, 0.1, 0, 1);
# Distrib (v,          TruncNormal          0.02,          0.01, 0.00000000000001, 100);
#
# # Calcul des vraisemblances
# Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2, 40, 2000);
Distrib (Fm,          TruncNormal_cv,          0.3,          0.2, 0.01, 1);
Distrib (r_pAm_pM,    TruncNormal_cv,          0.8,          0.2, 0.01, 1);
Distrib (kap,         Uniform,              0.1,          0.9);
Distrib (Ehp,          TruncNormal_cv,          100,          0.2, 40, 300);
# Distrib (E_coc,      LogUniform,              10,          300);
Distrib (rOM_ClxHorse, TruncNormal,          0.01,          0.2, 0, 1);
Distrib (kJ,          TruncNormal_cv,          0.003,          0.2, 0.00001, 1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

```

```

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIt)

# ./mcsim.DEBcaliS DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? TRUE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	133.98700	0.065239700	0.999582000	0.570799000	77.57670	12.945200
2.5%	121.57300	0.056237000	0.992249000	0.560703000	66.91500	10.658900
97.5%	143.60800	0.072550100	0.999948000	0.581188475	113.81100	14.353100
Min	71.18160	0.050052800	0.436289000	0.143151000	56.94700	10.001000
Max	154.22300	0.370152000	1.000000000	0.693264000	133.14000	260.208000
Result_mean	132.55446	0.063505182	0.997907584	0.570489568	84.51823	12.448501
Result_sd	5.54254	0.004658207	0.003613341	0.005748245	14.58372	1.570743

	rOM_ClxHorse.1.	kJ.1.
Result_mode	2.689640e-05	0.0026789200
2.5%	6.843740e-05	0.0015925500
97.5%	1.052460e-02	0.0039264700
Min	4.946710e-07	0.0003172570
Max	3.888110e-01	0.0051711200
Result_mean	3.315237e-03	0.0027719027
Result_sd	3.602102e-03	0.0006010289

Number of observations, n = 363

Number of parameters, k = 7

Log-likelihood, LL = -504.5171

BIC = 1050.295

Potential scale reduction factors:

	Point est.	Upper C.I.
E_coc.1.	1.02	1.04

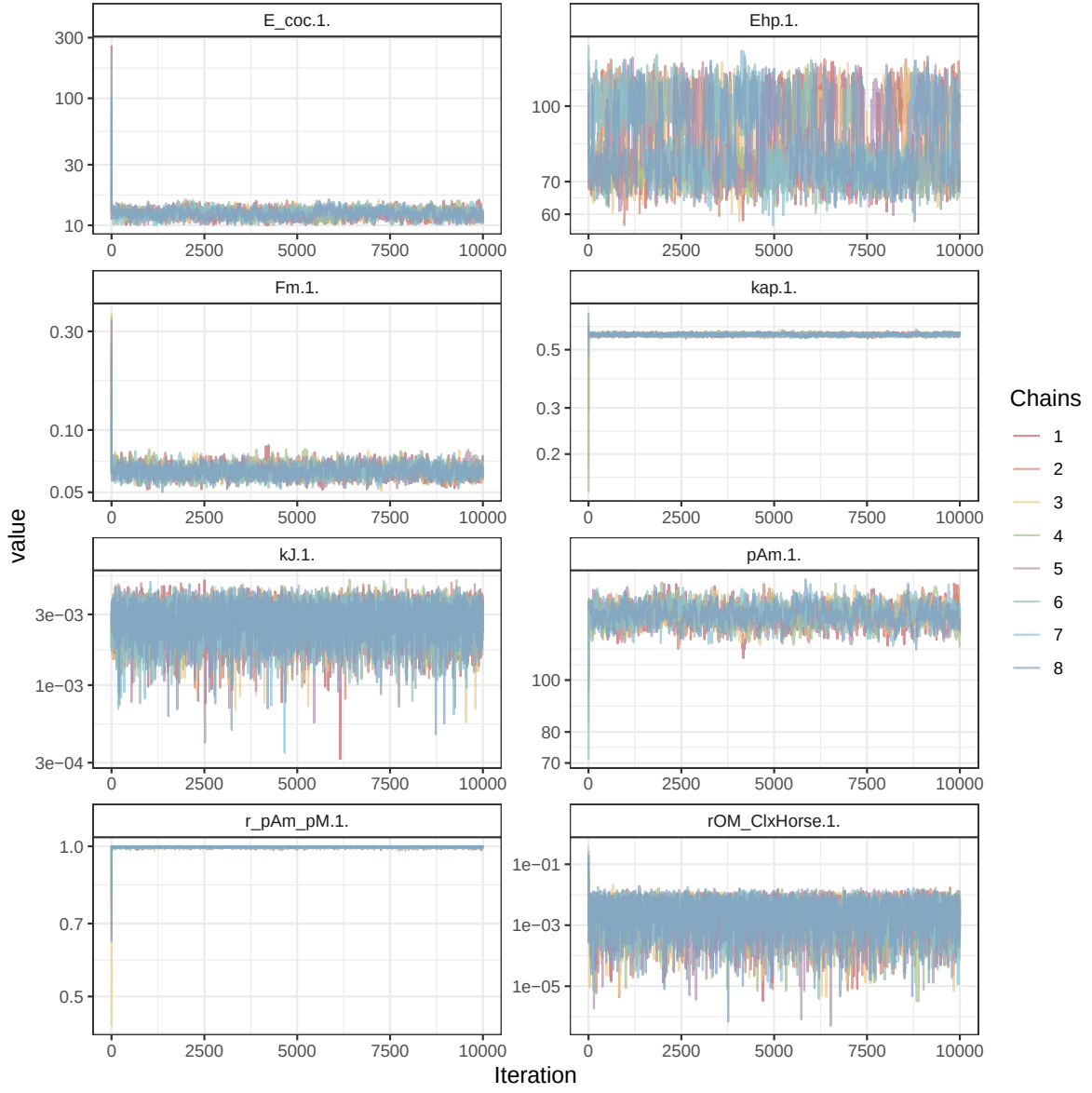


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

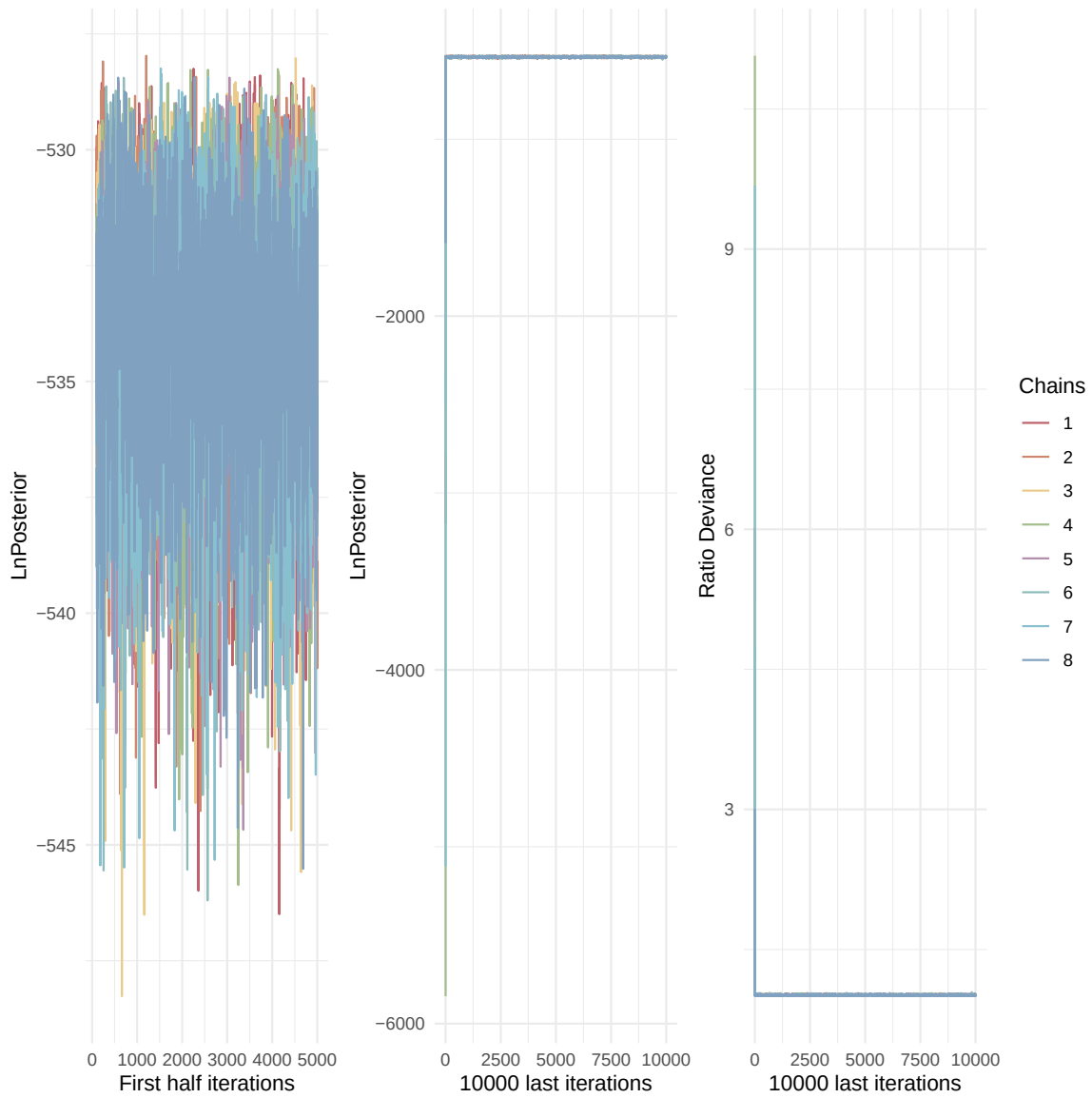


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

Ehp.1.	1.04	1.09
Fm.1.	1.01	1.03
kap.1.	1.01	1.02
kJ.1.	1.00	1.00
pAm.1.	1.01	1.02
r_pAm_pM.1.	1.00	1.00
rOM_ClxFhorse.1.	1.00	1.01

Multivariate psrf

1.04

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

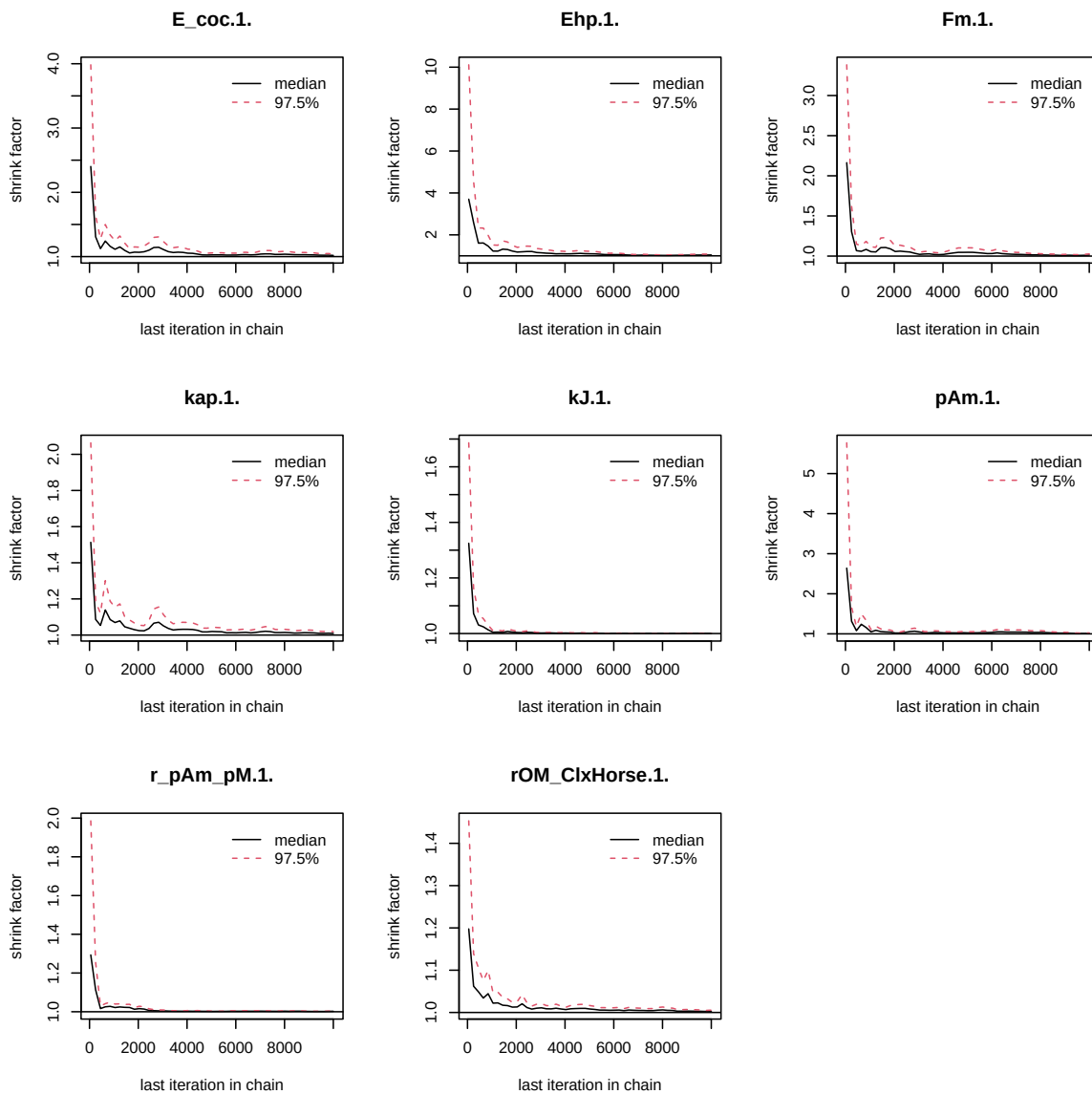


Figure 3: Chains convergence.

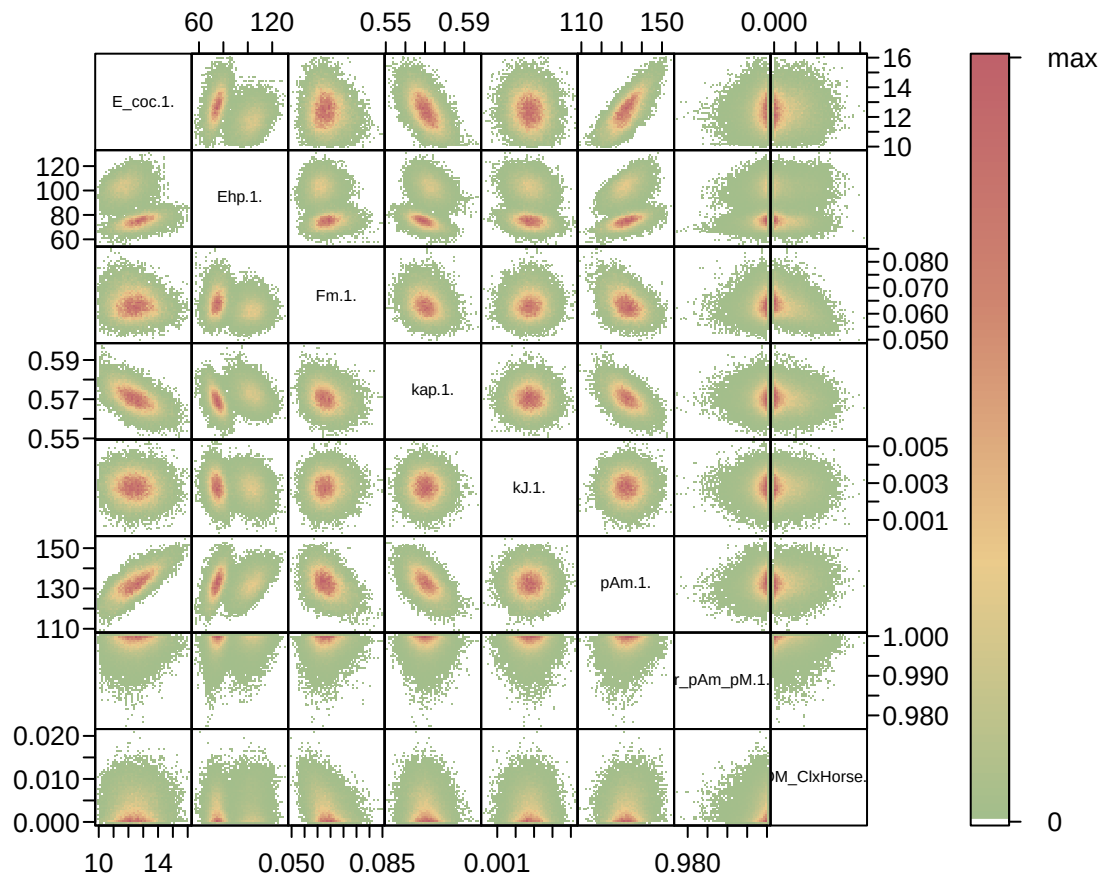


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```


**Posterior
and prior distributions**

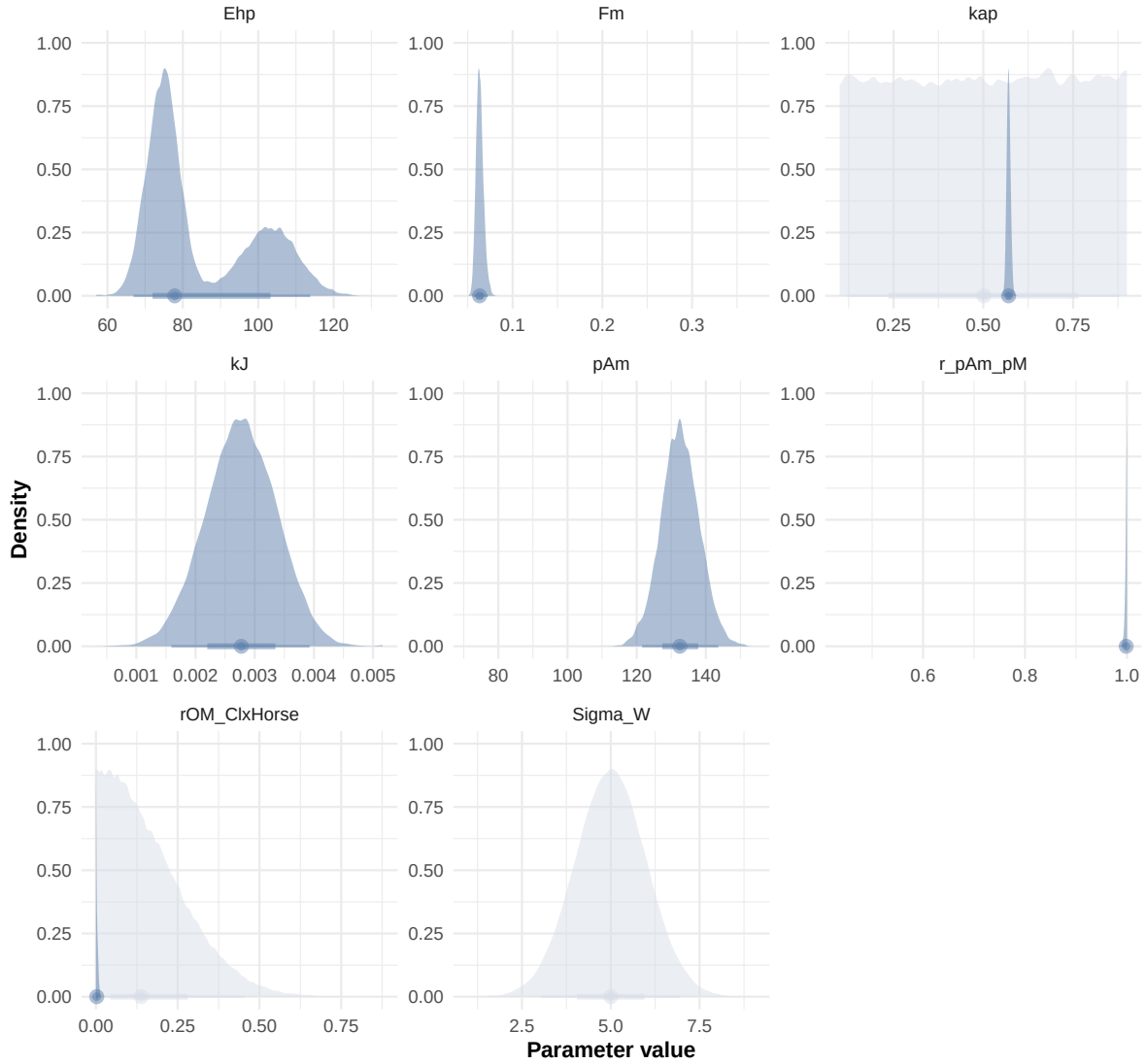


Figure 5: Distributions of the priors and the posteriors.

```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

# text_param <- "pAm = 1172;
# r_pAm_pM = 1.5;
# Fm = 0.5;
# Ehb = 0.5;
# Ehp = 20;
# kap = 0.55;
# kJ = 1.54032e-02;
# Eg = 6880;
# muOM = 12170.9;
# rOM_ClxFhorse = 0.3;
# v = 0.02;
# E_coc = 467;
# Sigma_W = 0.445952;"

# text_param <- "pAm = 1172;
# r_pAm_pM = 1.5;
# Ehb = 0.5;
# Ehp = 20;
# kap = 0.40;
# rOM_ClxFhorse = 0.3;
# E_coc = 467;
# Sigma_W = 0.445952;"

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%

```

```

mutate(
  param = gsub("\\.1\\.|$|\\.1\\.\"", "", param),
  line = glue("{param} = {val};")
) %>%
pull(line) %>%
paste(collapse = "\n")

# text_print <- "
#   PrintStep(freal,
#             kapreal,
#             Maturity,
#             Energy,
#             Organic_matter,
#             Weight,
#             Reproduction,
#             0, 70, 0.1);
#
# PrintStep(freal,
#           kapreal,
#           Maturity,
#           Energy,
#           Organic_matter,
#           Weight,
#           Reproduction,
#           70, 120, 0.01);
#
# PrintStep(freal,
#           kapreal,
#           Maturity,
#           Energy,
#           Organic_matter,
#           Weight,
#           Reproduction,
#           120, 400, 0.1);
# "

Endpoints_print <- "Weight,
Reproduction"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

```

```
# ./mcsim.DEBcaliS DEB_sim_full.in
```

```
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),  
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],  
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])  
)  
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)  
#####  
  
df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>  
  left_join(df_carc_exp, by="ID_experiment")  
  
# time_limits <- df_data %>%  
#   group_by(ID_experiment) %>%  
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)  
  
select_times <- seq(0,400, 0.1)  
  
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>  
  left_join(df_No_sim, by="No_sim") |>  
  left_join(df_carc_exp, by="ID_experiment") |>  
  # left_join(time_limits, by = "ID_experiment") |>  
  # filter(Time <= time_cutoff) |>  
  # dplyr::select(-time_cutoff) %>%  
  filter(Time %in% select_times)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexp
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
 "many-to-many"` to silence this warning.

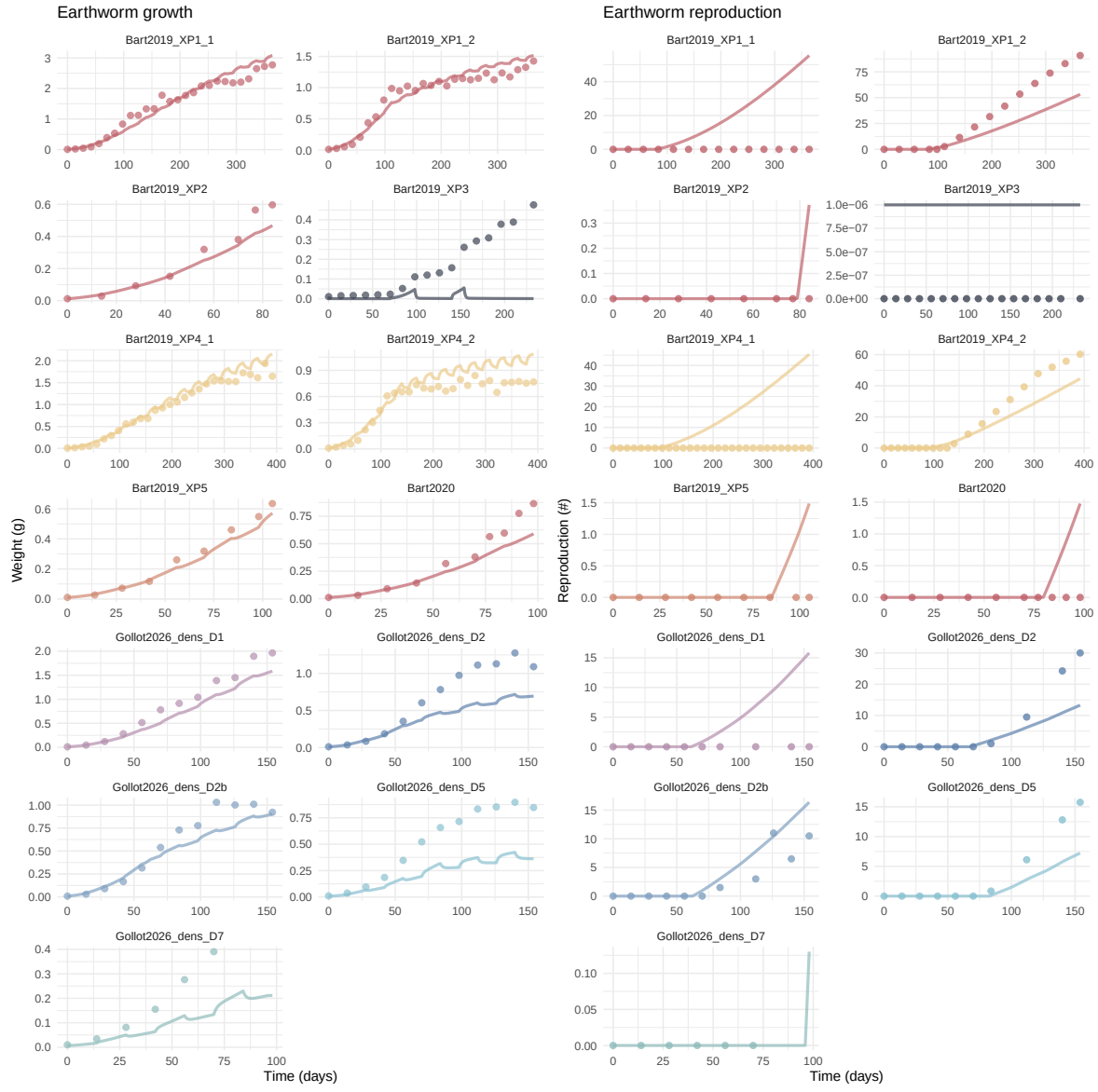
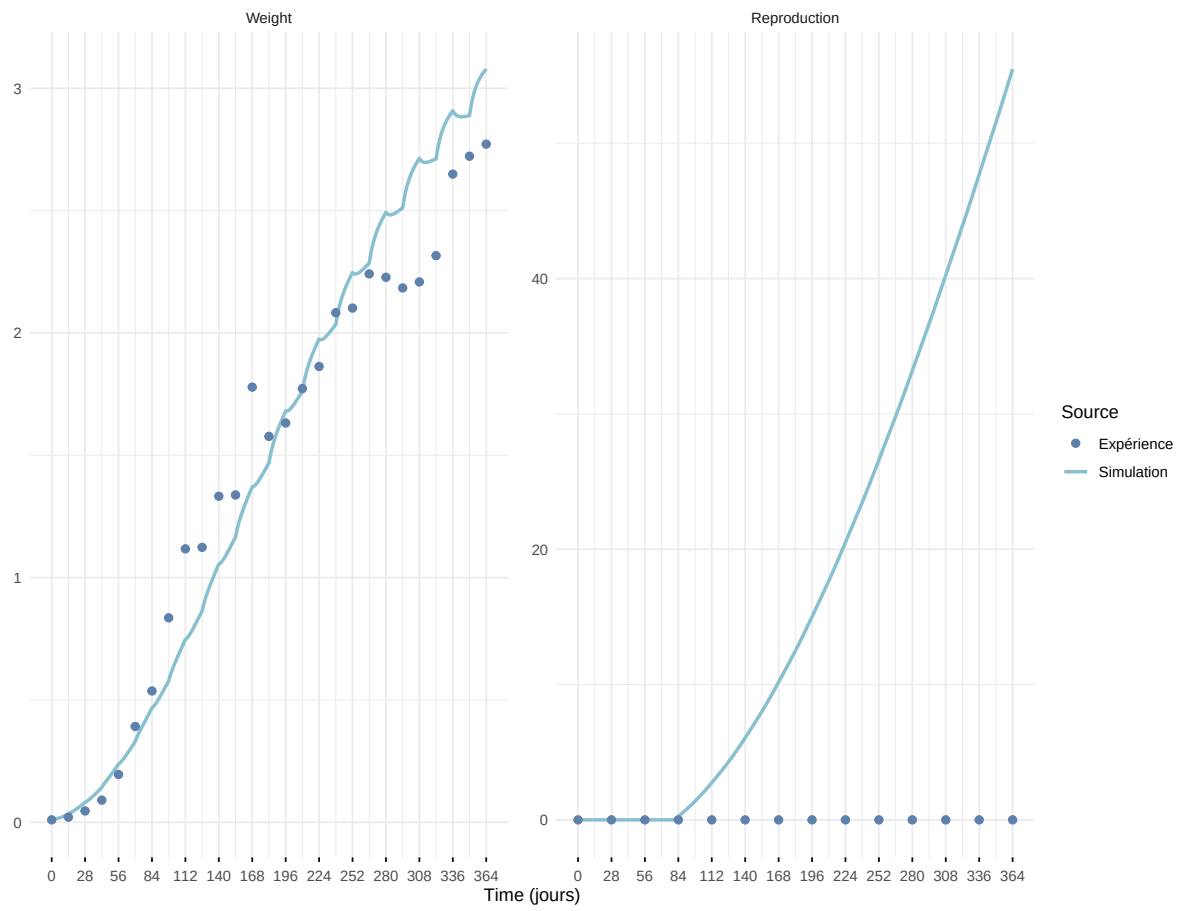
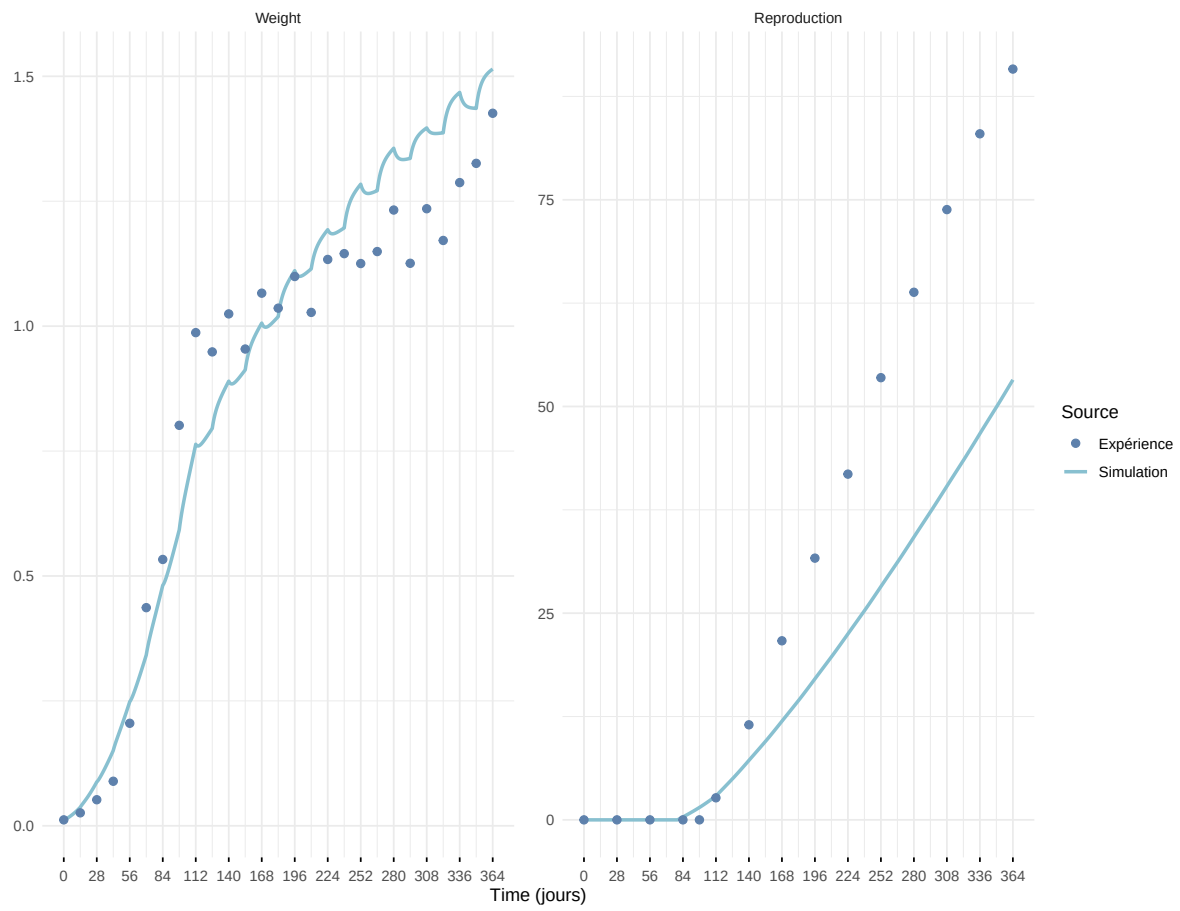


Figure 6: Simulations Weight and Reproduction.

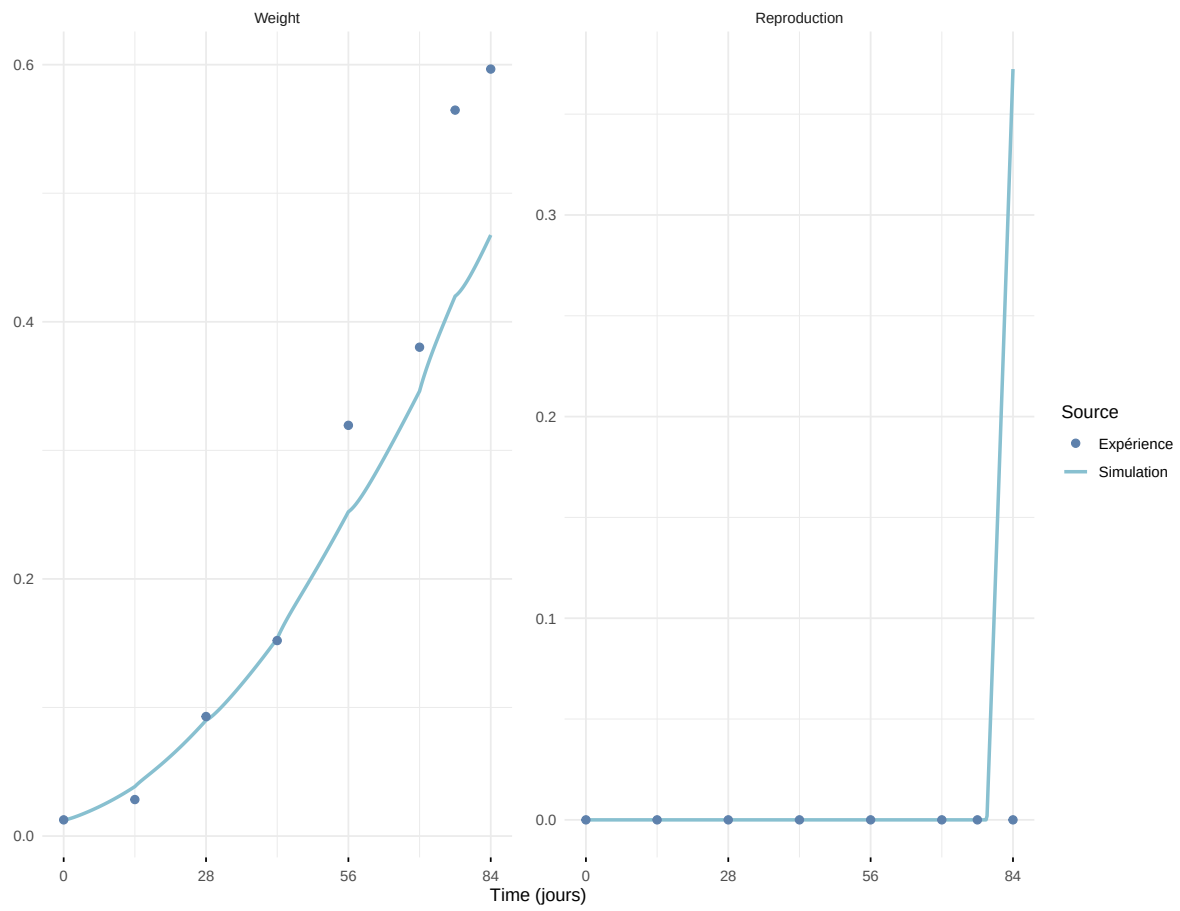
Expérience : Bart2019_XP1_1



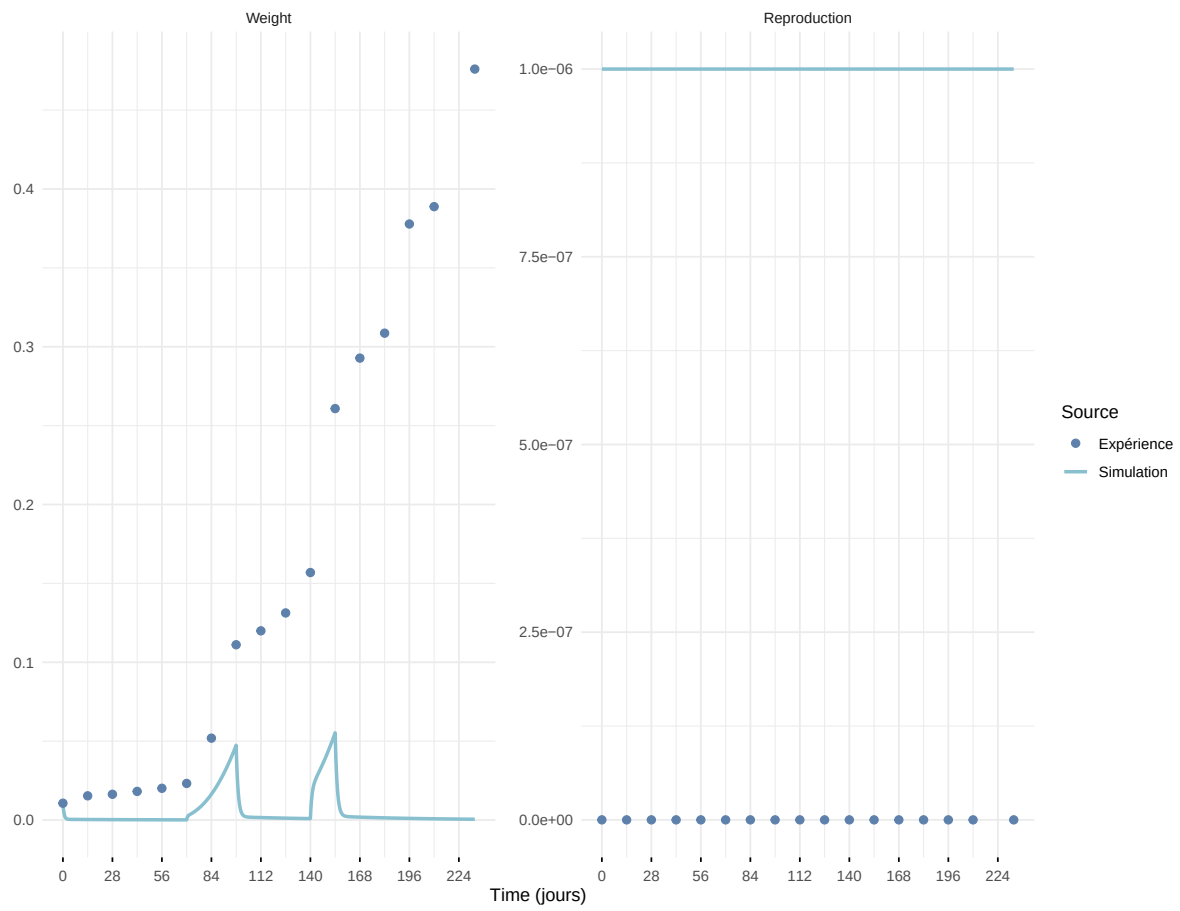
Expérience : Bart2019_XP1_2



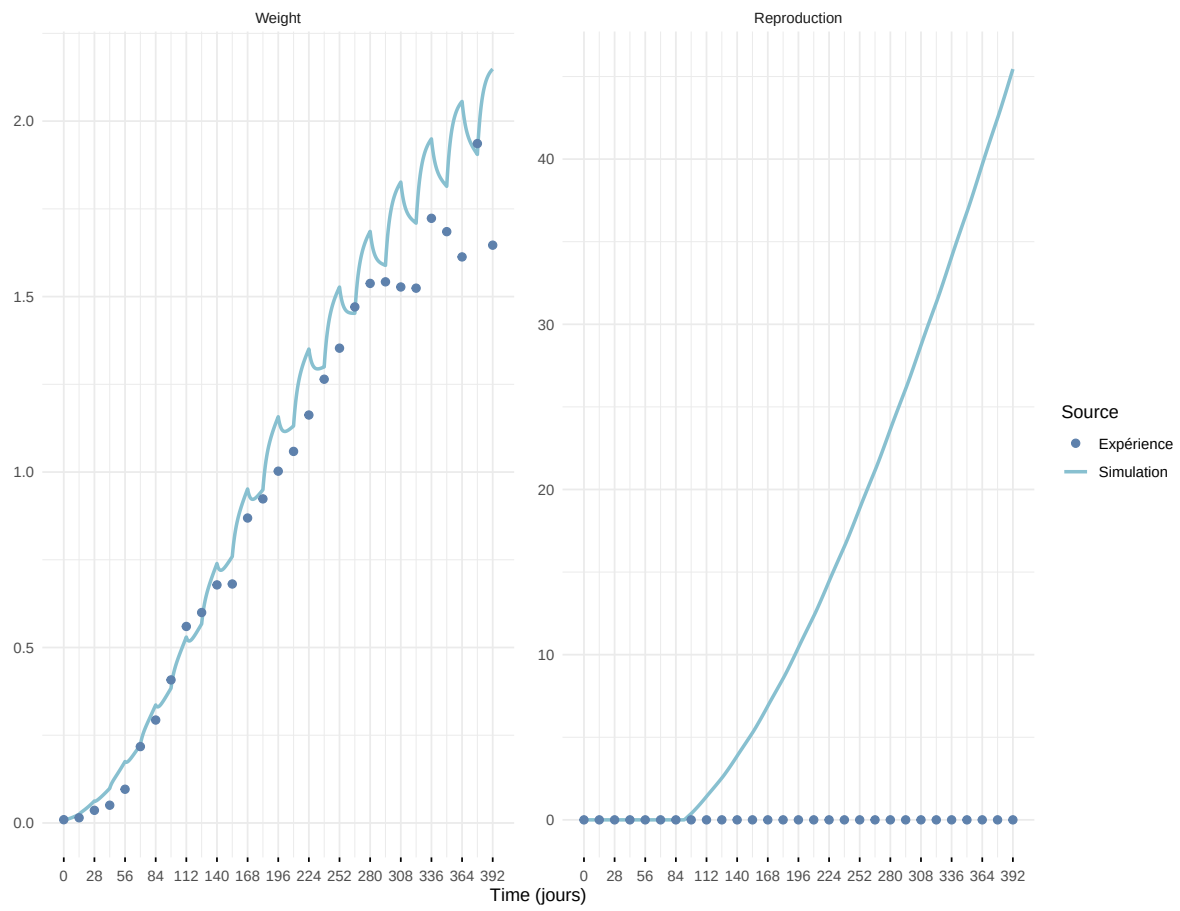
Expérience : Bart2019_XP2



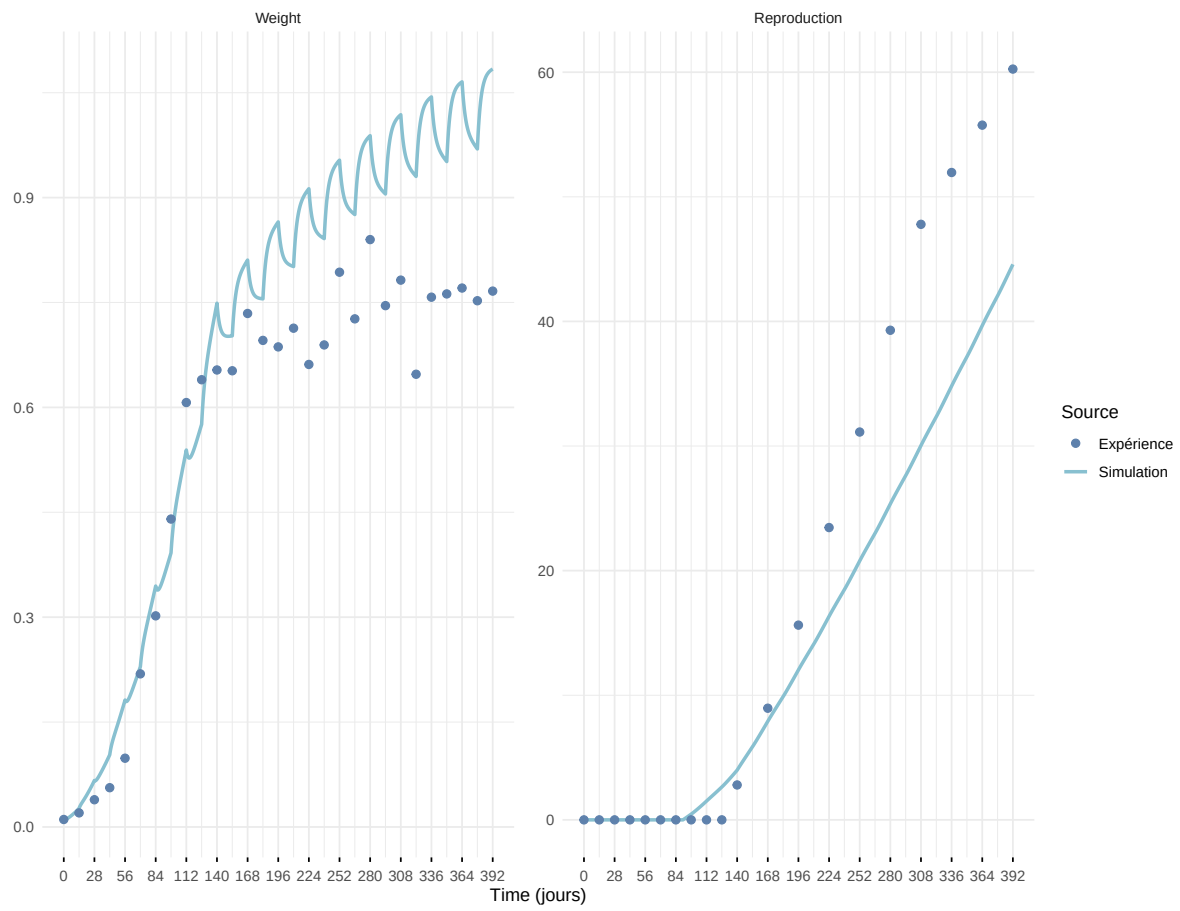
Expérience : Bart2019_XP3



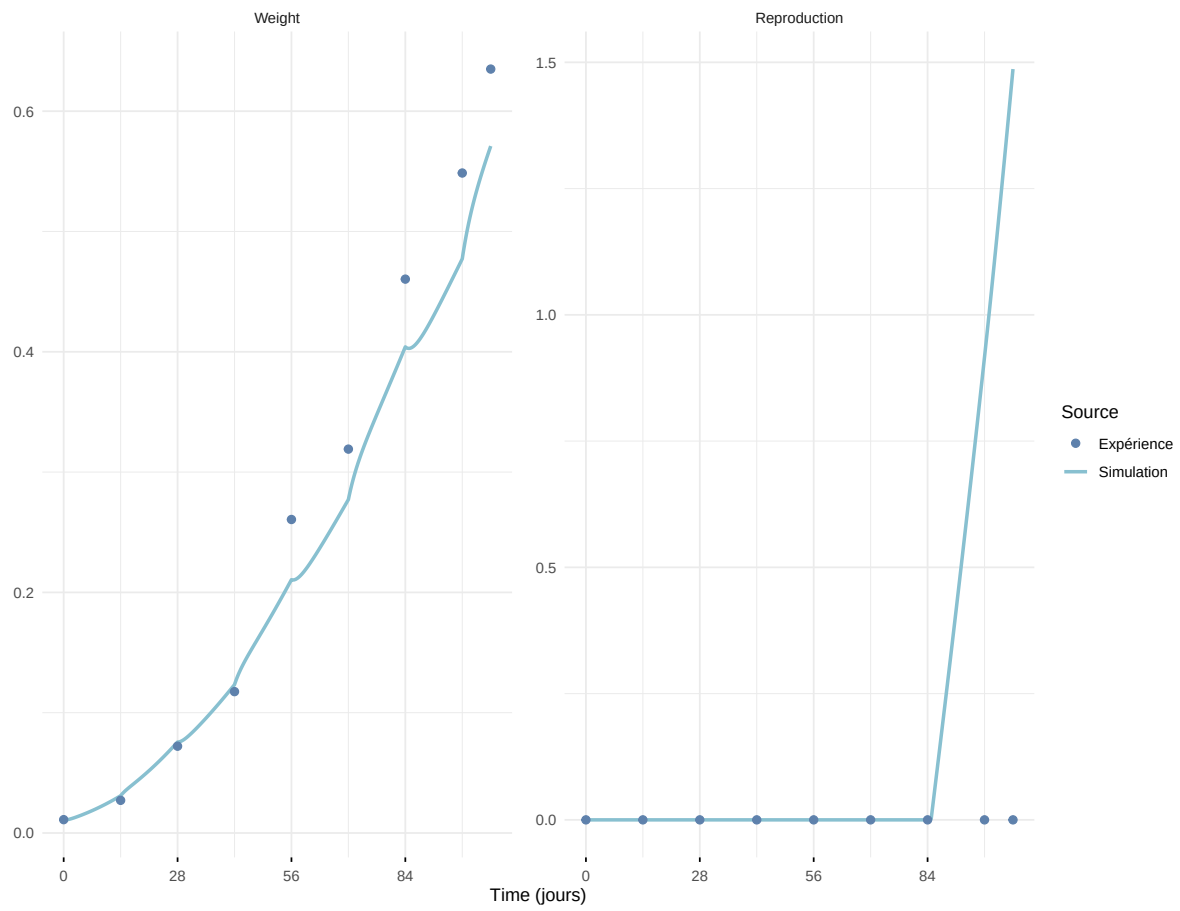
Expérience : Bart2019_XP4_1



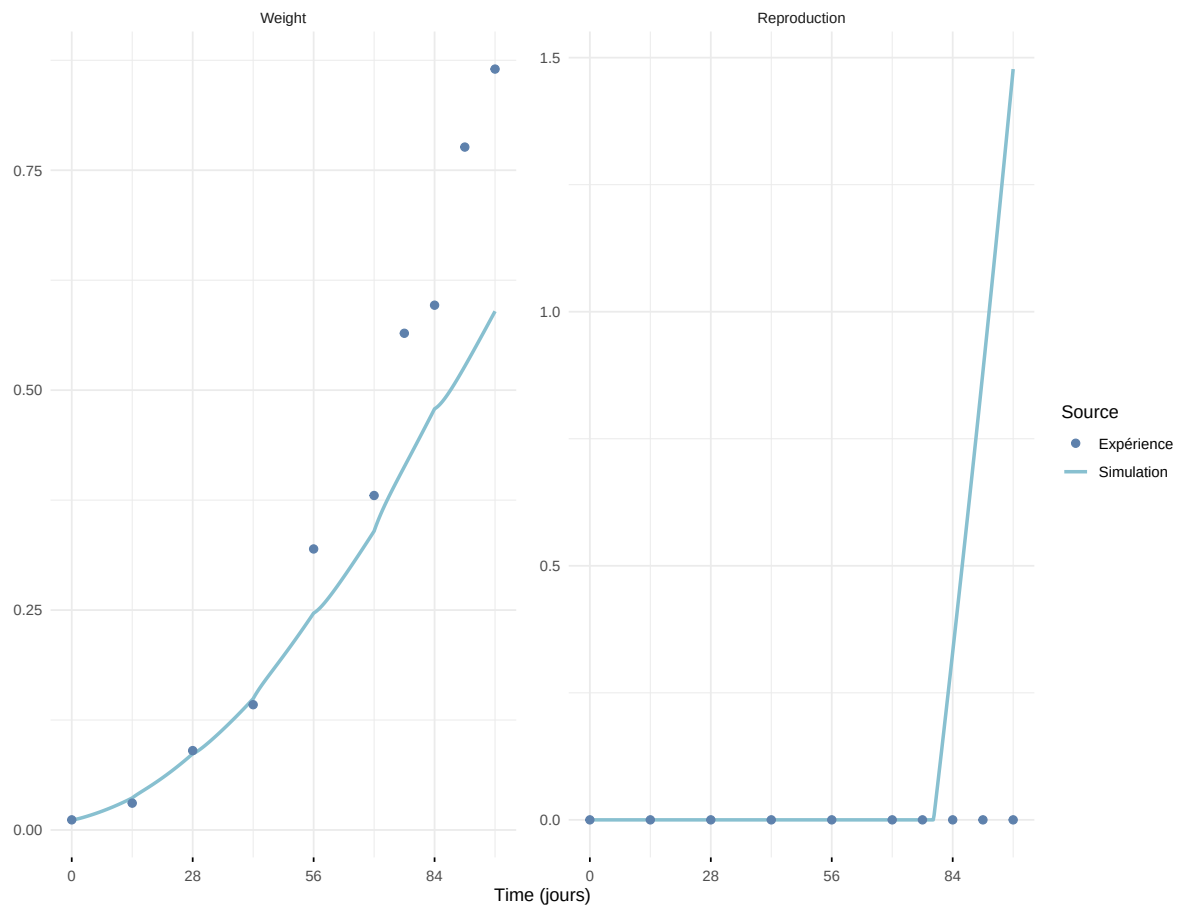
Expérience : Bart2019_XP4_2



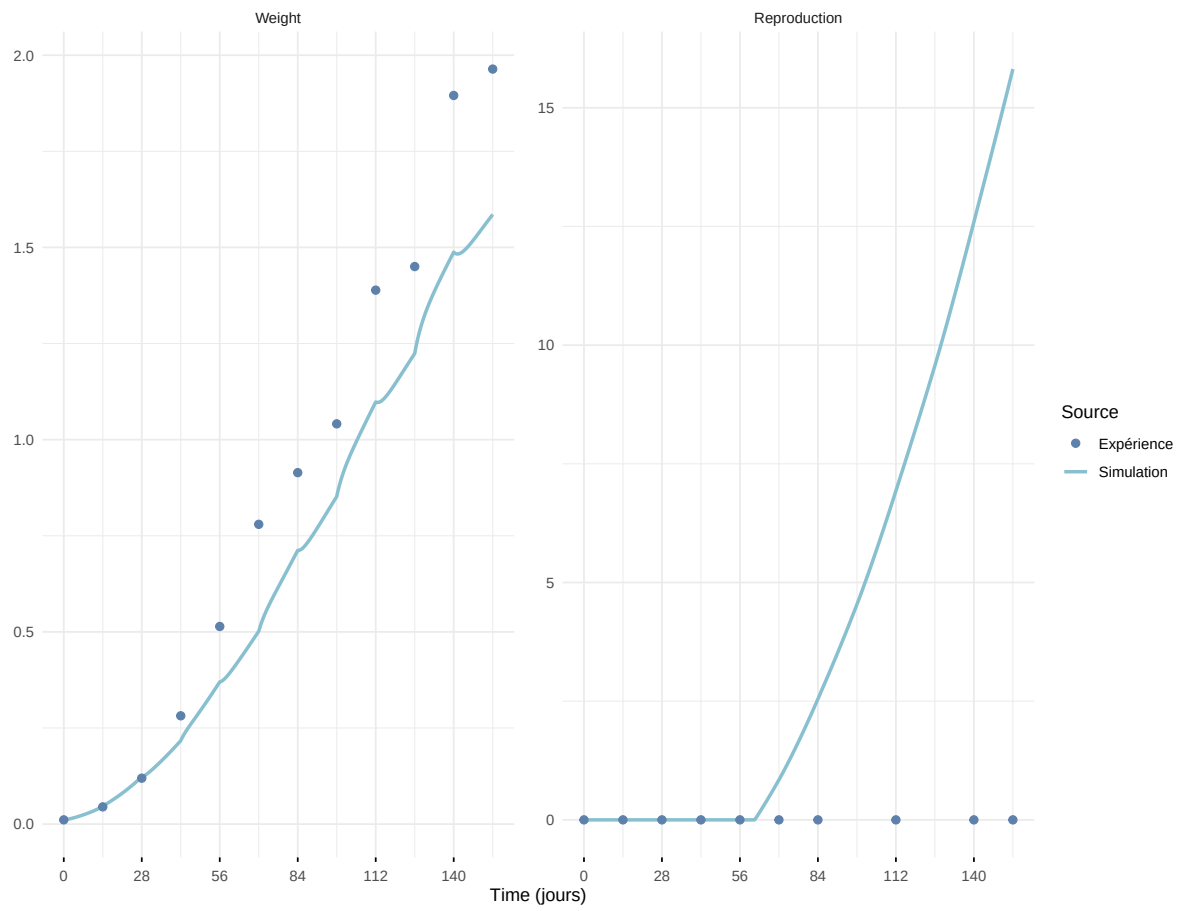
Expérience : Bart2019_XP5



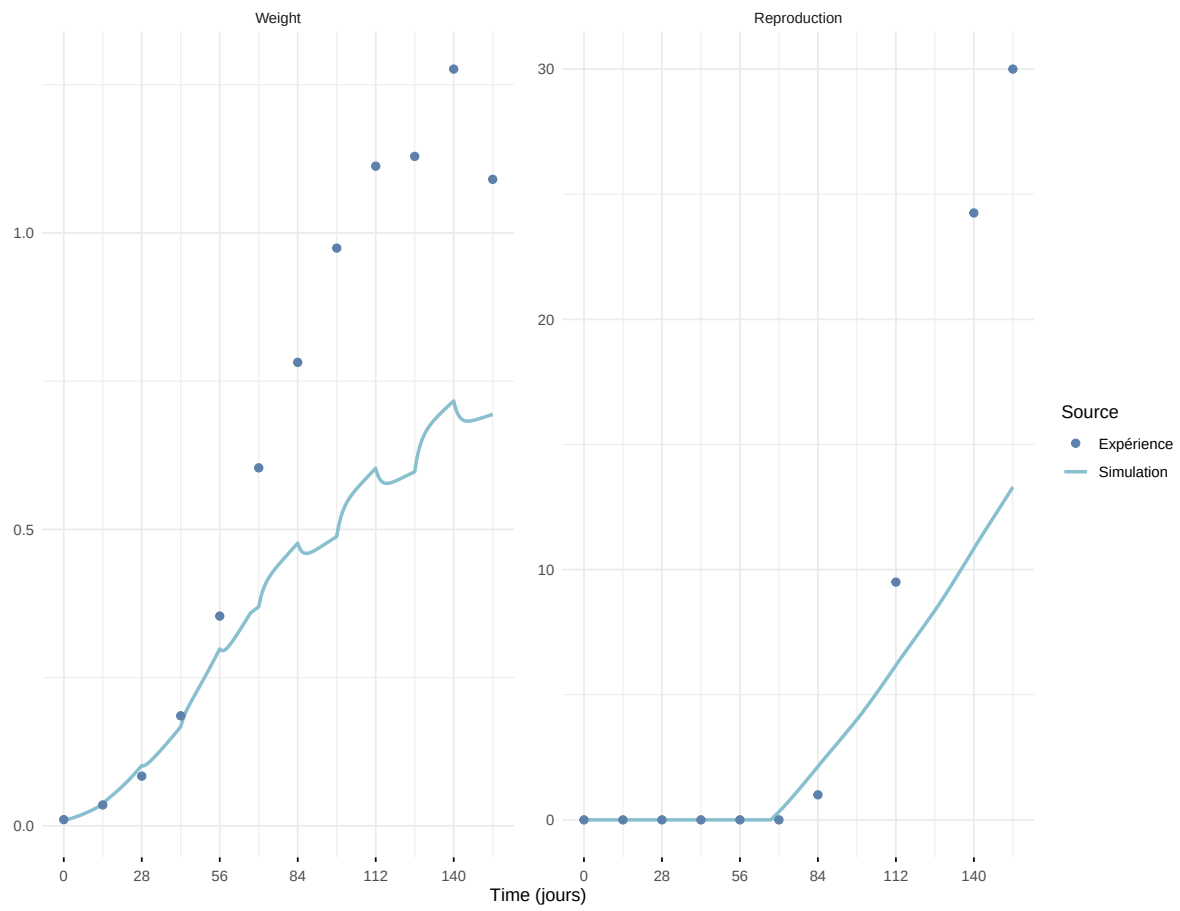
Expérience : Bart2020



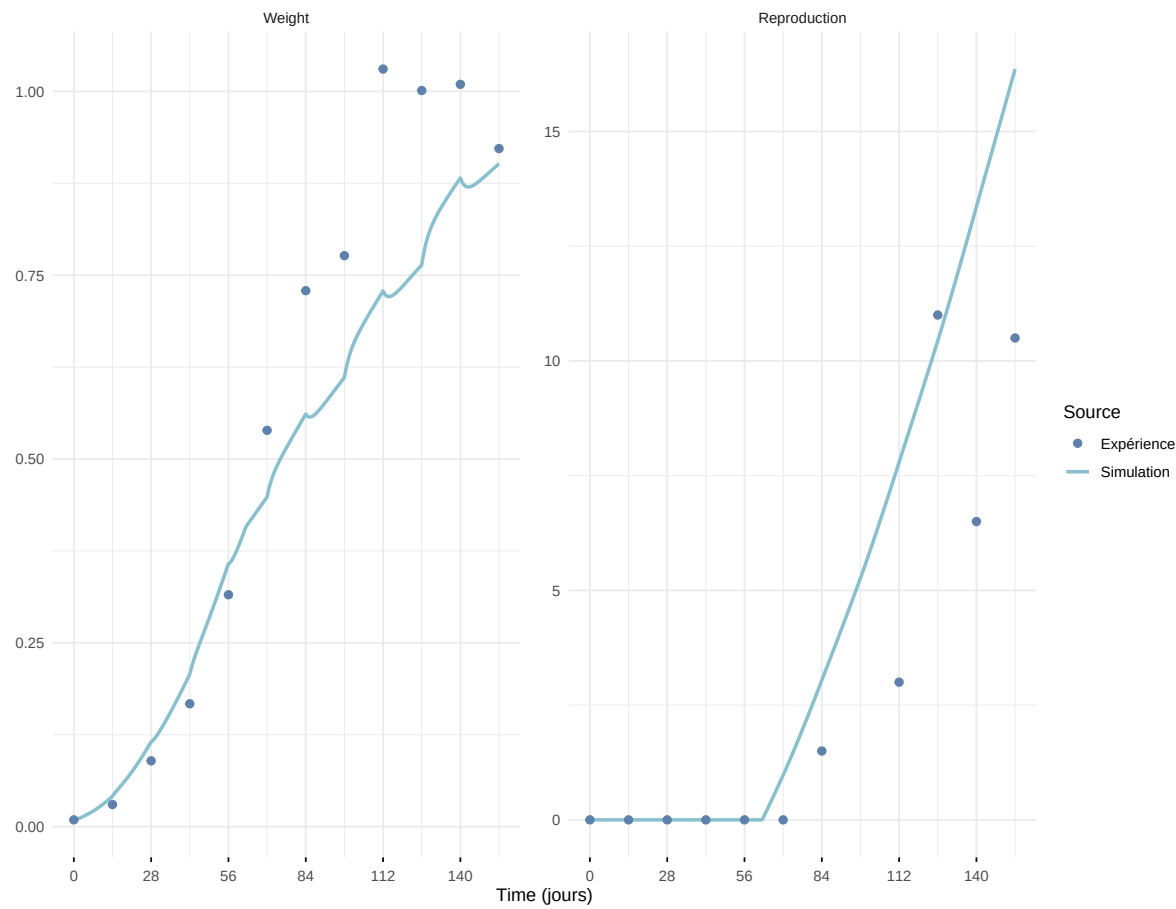
Expérience : Gollot2026_dens_D1



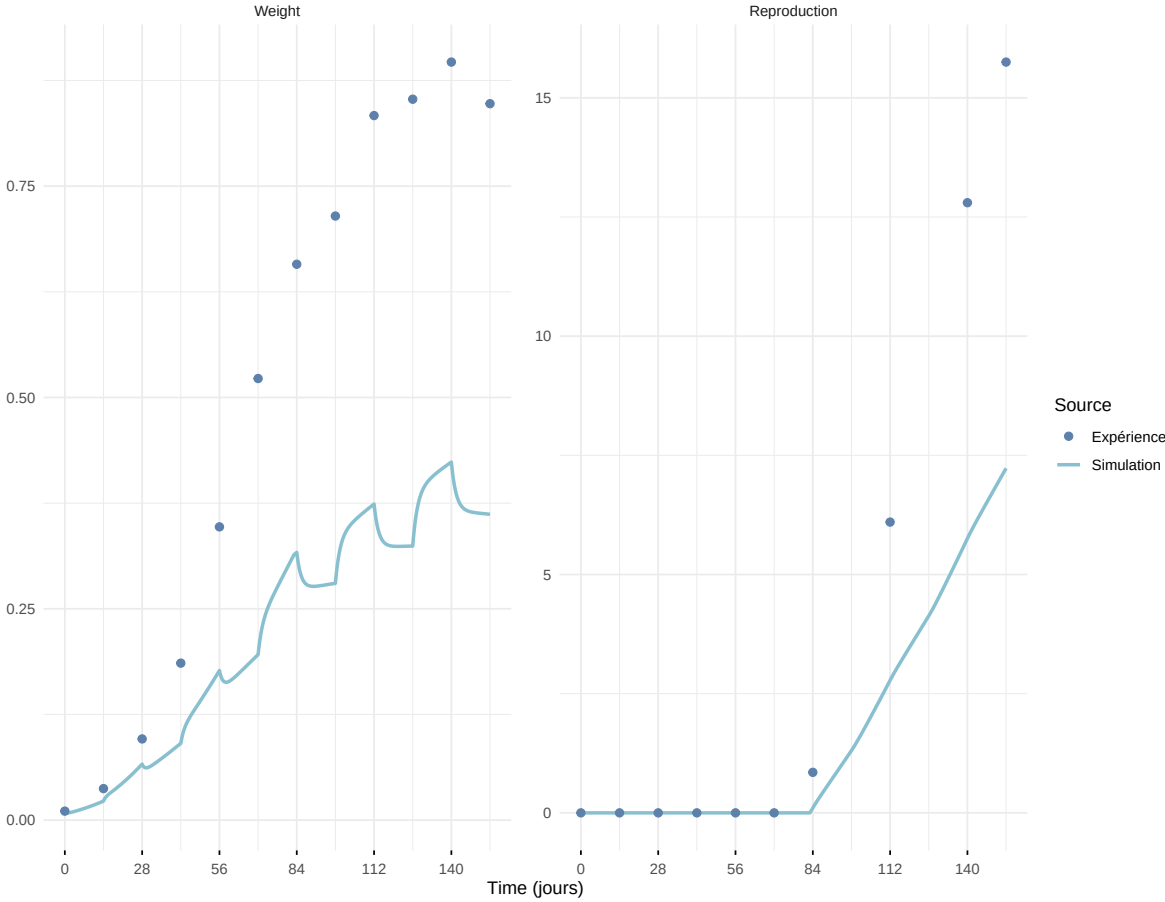
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

